# MTH 200 Chapter 4: Applications of Derivatives

## 4.2 Linear Approximations & Differentials Worksheet

1. Find the linearization of the function $f(x)=\sqrt{x+3}$ at $x=1$.

2. Find the linearization of the function $f(x)=\ln(2x-3)$ at $x=2$.

3. Use a linearization to approximate the number $\sqrt{3.98}$.

4. Approximate the number $(1.06)^6$.

5. Use the techniques of linear approximation to estimate the following quantities. Write each answer in decimal form. *Show all work by hand*.

a) $\ln(0.9)$

b) $\sqrt{104}$

c) $\sin\left(\frac{\pi}{4}+0.02\right)$

6. Consider the function $y=2x^2-\frac{1}{x}$.

   a) Approximate $\Delta y$ and $dy$ as $x$ changes from 1 to 2.

   b) Sketch a picture of the curve and the values of $\Delta y$ and $dy$.

7. Consider the function $y = x\sqrt{8x+1}$.

a) Find general formulas for $\Delta y$ and $dy$.

b) Approximate $\Delta y$ and $dy$ when $x = 3$ and $\Delta x = 0.01$.